



Girls in Security Operations (GiSOC) Bootcamp

MAMBA RANSOMWARE ANALYSIS

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1. Executive Summary

The sample 131.exe was analyzed to understand how the Mamba ransomware family operates. Mamba performs full disk encryption using DiskCryptor components rather than encrypting files. Its activity leads to complete system lockout after a reboot.

During the investigation, the malware installed DiskCryptor drivers, modified registry keys, created supporting processes, and attempted outbound network activity. The hashes of the file were submitted to VirusTotal and flagged as suspicious by most engines.

The main risk is total loss of system access. Recovery is only possible through offline backups or system rebuilds.

2. Overview

Mamba ransomware targets system volumes and replaces normal boot behavior with encrypted disk access. The sample uses DiskCryptor components to take control during the next reboot. Once active, the malware blocks the operating system from loading.

The execution chain includes writing drivers to disk, creating processes linked to DiskCryptor, and modifying system settings to prepare for encryption. The observed behavior matches documented cases of Mamba attacks.

3. Scope

The scope of this report includes:

- Classification of the sample
- Static review of embedded resources, imports, and hashes
- Dynamic review of process, registry, and network activity and Assessment of overall impact

4. Environment

The analysis was completed in an isolated lab using FlareVM and Remnux (Host-only network) and Inetsim for controlled DNS and HTTP responses No access to external networks.

This setup ensured safe execution without risk to production systems.

5. Tools Used

Static Analysis Tools: PEStudio, Strings, Virus Total and hashmyfile (SHA256, MD5 generators)

Dynamic Analysis Tools: Regshot, Process Monitor, Wireshark and remnux

These tools were used to capture registry changes, file operations, process creation, and network activity.

6. Analysis and Investigation

6.1 Static Analysis

File Information - SHA256:

2ecc525177ed52c74ddaaacd47ad513450e85c01f2616bf179be5b576164bf63 - MD5:
409d80b694643fbc4a1fa61c07806883

- Size: 2.30 MB
- Architecture: 32-bit
- Windows executable

VirusTotal flagged both hashes as suspicious. Almost all security vendors classified the file as ransomware or a Disk Cryptor based threat.

The screenshot displays the VirusTotal analysis page for a file named 131.exe. The file's SHA256 hash is 2ecc525177ed52c74ddaaacd47ad513450e85c01f2616bf179be5b576164bf63. The file is identified as a Windows executable (EXE) with a size of 2.30 MB. The analysis shows that 66 out of 71 security vendors flagged the file as malicious. The file is associated with several threat labels, including trojan.dcryptor/hddcrypt, ransomware, and hddcrypt. A table of security vendors' analysis is also visible, showing detections from AhnLab-V3, AlCloud, and Avira.

Security vendors' analysis	Do you want to automate checks?
AhnLab-V3	Malware/Win32_RL_Generic.R371154
Alibaba	Ransom:Win32/Blocker.0720e0b
AlCloud	RansomWare
Avira	Trojan.Ransom.HDDCryptor
Avira	Trojan.Ransom.HDDCryptor

Fig6.1_virustotal

SOC-Malware-001



Strings showed disk-related terms and references to mounting and driver operations, consistent with disk encryption behavior.

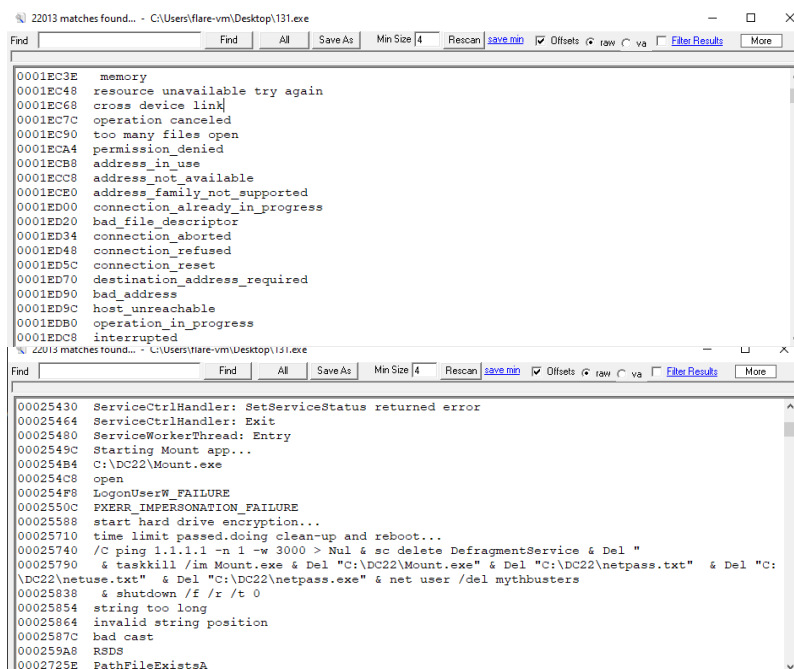
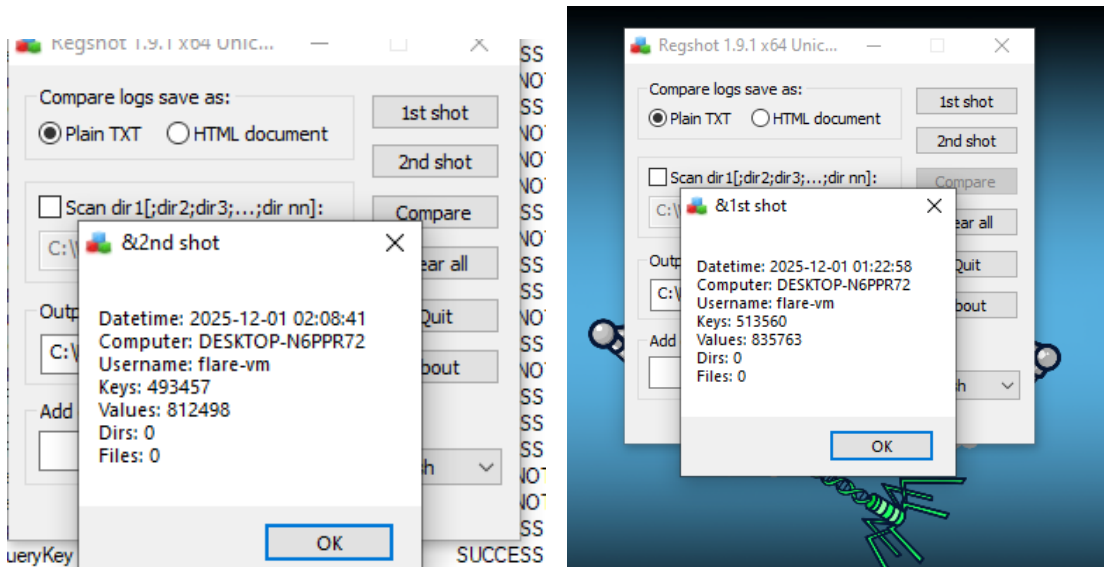


Fig6.1.2_Strings

6.2 Dynamic Analysis

Registry Changes The malware changed more than one hundred registry values. It removed thousands of existing keys and created new ones linked to DiskCryptor and boot configuration. These changes prepare the system for encryption.



Processes and File Activity The sample created new processes tied to the embedded DiskCryptor executables. It wrote drivers and support files to disk. System boot behavior was altered through registry modifications and new services.



The image shows the Wireshark network protocol analyzer interface. The top menu bar includes File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, and Help. Below the menu is a toolbar with various icons for file operations, network analysis, and search. The address bar shows the URL 'http'. The main packet list displays a series of HTTP GET requests to '/msdownload/update/v3' from source IP 10.0.0.4 to destination IP 10.0.0.3. The selected packet (Frame 19) is expanded in the packet details pane, showing the following structure:

- Frame 19: 342 bytes on wire (2736 bits), 342 bytes captured (2736 bits) on interface enp0s17, id 0
- Ethernet II, Src: 08:00:27:f3:2e:5a (08:00:27:f3:2e:5a), Dst: 08:00:27:ca:9c:c5 (08:00:27:ca:9c:c5)
- Internet Protocol Version 4, Src: 10.0.0.4, Dst: 10.0.0.3
- Transmission Control Protocol, Src Port: 49672, Dst Port: 80, Seq: 1, Ack: 1, Len: 288

7. Conclusion

Mamba ransomware is a high impact threat that locks entire system volumes. The sample analyzed follows the established behavior of this family by installing DiskCryptor drivers and changing boot settings. VirusTotal results confirm widespread detection, and dynamic analysis shows active preparation for full disk encryption.

If executed on a live system with administrative rights, the malware would make the system unusable after reboot.

8. Recommendations

- Monitor driver installations and suspicious service creation
- Track changes to boot configuration and registry autoruns
- Use endpoint tools that detect DiskCryptor abuse
- Maintain offline backups and verify recovery procedures

9. Lessons Learned

- Early detection relies on spotting driver drops and boot changes
- Safe malware analysis requires a controlled and isolated environment
- Recovery is only possible with complete backup strategies